



The Snap

Artist's Proof 44

Forces and Constants

The configuration cost — the arena's once-held commitment that realises the gravitational coupling

Artist's Proof 44 — LOCKED 2026-08-11 at composite 9.3 under Reviewer Brief v1.1 (weakest link: derivation rigour — §3 at the structural register with D-CCC.1 declared, on the AP30/KS-30.1 precedent).

Title ruled: The Snap. Provenance: PROV-CCC.1 through PROV-CCC.7, the two session records, the transcript exports, and the analog archive, published beside this proof.

Artist's Note

What this paper does — and why.

This paper names and prices the cost the corpus has already admitted to. AP06 priced the flow: the leakage, one unpaired ε shed at every instant, bounded by c , the constant of every leakage denominator. AP28 named a second cost without pricing it — “Gravity is the cost the arena pays to keep the break open” — and then derived G as though the arena paid nothing to be constituted. This paper prices the constitution. The price is the coupling configuration cost — the snap cost, CCC — and it is the axiom's own arithmetic: one unpaired element, held once, at the one snap that matters: the arena's own. The realised gravitational coupling is the structural coupling divided by $(1 + \varepsilon)$.

The argument in one breath. Every Actualization State instant produces exactly one unpaired ε (AP43 D5, carried by AP06 §6). At every instant after a configuration exists, that instant's ε is shed at the boundary: the leakage. At a constitution instant the ε cannot be shed — if it were, nothing present would hold the first record open, and by Axiom R the crack would heal and the configuration would not persist. Therefore it is held. Applied where the corpus's own clause applies it — “the crack is open everywhere or nowhere” — the constitution that concerns a universal constant is the arena's: one arena, one snap, one ε . Matter configurations hold constitution commitments of their own, but theirs are already booked: by AP29, mass is held coupling capacity actualised, so a configuration's commitment is a constituent of its measured mass, not a correction to it. The arena's commitment is the single unbooked one, and it lives in the arena's coupling.

What it touches and what it cannot touch. No mass is dressed, so every mass ratio is invariant trivially: the proton result of AP30 — inside experimental uncertainty — is preserved to the digit, and AP30 §7's reading of its own residual stands unamended. The commitment appears in exactly one place: the coupling the corpus has already named as the arena's cost. The realised prediction is $G = \alpha_{em}^{21} \times (1 + 1/\pi) / (1 + \alpha_{em}) \times \hbar c / m_e^2 = 6.6719 \times 10^{-11} \text{ N}\cdot\text{m}^2/\text{kg}^2$ — closing about ninety-five per cent of AP28's 0.69% and moving the prediction from outside every laboratory determination ever made to the low edge of the measured landscape.

The honesty box. The load-bearing proposition — held, not shed, at constitution — is derived here at the structural register, with its record-algebra formalisation named as a debt, not hidden. The value of ε in the quotient is a selection among the domain readings AP06 holds open, argued in §5 and booked as a debt, not

smuggled as a derivation. The residual that remains after the commitment, about -0.036% , flips sign: it can no longer be absorbed by downward apparatus leakage, and it is reported open and unattributed. And the landing itself is weak evidence by the paper's own §10: the paper is worth having for the structure and the falsification surface, not for the number. Seven kill switches engage, and the first is aimed at the claim, not the author: KS-CCC.1 fires if any refinement of the commitment stands in canon whose selection depended on comparison with measurement and whose history is undisclosed. Two clean fractions the data volunteered after the fact — $20/21$ of ε and $17/18$ of ε — are named in §10 for exactly one purpose: to be fenced. Seen, and refused.

The completeness claim. This paper adds no new axiom, no new measured input, and no new number. It reads a sentence the corpus locked years ago and prices it with the element the axiom already provides, at the instant the axiom first provided it. The description was incomplete by one commitment. Now it is written down.

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Orientation

Where this paper fits

This paper is Artist's Proof 44, shelved in Notebook IV (Forces and Constants) immediately after AP28 (The Constant), whose result it completes and — on adoption — amends. It depends on Notebook I for the axiom $1:1 + 1 \times \varepsilon$ @ AS and the four conditions {S, B, R, C}; on AP06 for the leakage structure and Identification 6.1; on AP28 for the coupling graph, the per-channel coupling, the R-reading of G, and the everywhere-or-nowhere clause; on AP24 for the many-readings principle §7 extends; on AP10 for the three dimensions and the identification of the 1:1 as the substrate; on AP43 D5 for the one- ε -per-instant account; on AP29 for mass as coupling capacity actualised — the result that books every matter configuration's commitment before this paper begins; and on AP20 for the unconditional floor.

Downstream, on adoption only: the wording of KS-R.7; AP28 §6–§7; Ø Predictions Part II, where this paper enters as a third candidate for the 0.69% and reinterprets the Appendix C proposal as a discriminator; the G passages of the Proofs and Prose registers; and every corpus locus that consumes the predicted value of G or of the holding limit κ numerically — formulas and values both — per the widened audit of D-CCC.3.

Reading order within this paper

Front to back. Sections 2 through 5 are the spine: the constitution record, the held commitment, the arena's single snap, and the form of the correction. Section 3 carries the load. Section 6 places the commitment — forced, under the arena reading. Section 7 extends a licence and says so. Section 8 is architecture — the ten vertices — and nothing numerical rests on it. Sections 9 through 11 are empirical: the prediction, the discipline that fences it, and its standing among the candidates Ø Predictions holds open. If you are reading for the result only, §5 and §9. If you are reading for rigour, §3. If you are reading to check that the author has not fooled himself, §10.

How to read this paper

Two registers alternate as the material requires, as everywhere in Notebook IV. Formal blocks open with a bold label (Proposition, Corollary, Kill Switch) and close with the mark ■. The connective passages speak directly. One register discipline

specific to this paper: the number enters the argument once, in §9, after the structure is complete, and §10 exists to keep it that way.

A note on notation

α_G is AP28's structural gravitational coupling, $\alpha_{em}^{21} \times (1 + 1/\pi)$: the provisioned coupling. α_G^* is the realised coupling, $\alpha_G / (1 + \varepsilon)$, introduced here. κ is the holding limit; $\kappa^* = \alpha_G^* \hbar c / 2$ is its realised value. ε carries one structural meaning throughout the corpus — the unique unpaired element of Axiom B. Its value in the quotient, $\alpha_{em} \approx 1/137$, is a selection among the domain readings AP06 holds open, argued in §5 and booked as D-CCC.4. “Snap” means a constitution event: the single Actualization State instant at which a configuration's first record is written and holds. “The Snap”, capitalised, means the arena's own: the original break, the first record of all. “Provisioned” and “realised” name the two totals the axiom itself distinguishes: $1:1 + 1 \times \varepsilon$, and the $1:1$ that persists. CCC is the arena's once-held commitment; TLC is the per-instant leakage of AP06. “Vertex” and “edge” refer to the coupling graph AP28 Propositions 1–2 prove: ten vertices, twenty-one edges.

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0 — Dependency and Scope

0.1 — What this paper does

This paper derives the coupling configuration cost — the once-held commitment paid at the arena's constitution — from the four conditions {S, B, R, C} and results already locked in the corpus. It establishes the constitution record (§2), proves at the structural register that a constitution instant's unpaired ε is held rather than shed (§3), fixes the count at one — one arena, one snap, one ε — and shows that matter's own commitments are already booked as mass (§4), reads the form of the correction off the axiom and argues the selection of its value (§5), places the commitment where only it can go (§6), states plainly that §7 extends a licence, and states the realised prediction once (§9).

What the paper does not do. It does not attribute the residual that remains after the commitment (§0.7). It does not derive any refinement of the magnitude below one ε , and §10 forbids selecting one empirically. It does not derive the boundary at which the commitment's value is read; §5 argues the selection and D-CCC.4 books it. It does not decide the vertex distribution (§8), on which nothing numerical rests. And it does not rest its worth on §9: by §10's own argument the landing is weak evidence, and the paper's worth is the structure and the falsification surface — Propositions 1 through 6 and the switches of §12 — stated here, before anyone reaches the number.

0.2 — Dependencies

AP06 (The Leakage Constant). Identification 6.1 and the open family of domain readings of ε (§0.7 there); the leakage as the shed ε . Load-bearing for §3's Step 5. §5's selection runs on AP28 Proposition 5 alone; AP06 supplies the open domain family the selection is made within — it constrains the field without grounding the choice, and the paper no longer claims otherwise.

AP28 (The Constant). The coupling graph (Propositions 1–2); the per-channel coupling (Proposition 5 — the element's one-channel coupling worth is α_{em} by definition); the R-reading of G (“the cost the arena pays to keep the break open”); the everywhere-or-nowhere clause of §4, which is arena-scoped in its own words. Load-bearing for §§3, 4, 5, 6, 8.

AP24 (The Residual). One object, many independent readings — the principle §7 extends, with the extension named. AP24 was locked before this paper existed.

AP10 (The Dimension). Three spatial dimensions, and §7's identification of the 1:1 as the substrate — the exhaustiveness step of §3.

AP43 D5. One unpaired ε per Actualization State instant. Load-bearing for §3 and §4.

AP29 (The Actualization Proof). $E \equiv \kappa(B) \cdot AS$ — mass as coupling capacity actualised. Load-bearing twice: it books every matter configuration's commitment as a constituent of its measured mass (§4, Corollary), and it corroborates the placement (§6).

AP20 (The Proof). Independence of {S, B, R, C}; the unconditional floor. This paper is unconditional on EH and QRA.

0.3 — Axiom mapping

Axiom S. The 1:1 — what persists as the realised arena. The balance is what The Snap delivered.

Axiom B. The $1 \times \varepsilon$ — the element committed. The break supplies the only currency the constitution instant has.

Axiom R. The holding — once, forever. The commitment is R's price, as the leakage is C's. This paper stands to Axiom R exactly as AP06 stands to Axiom C.

Axiom C. The boundary structure at which readings of ε are taken. C prices the flow; R prices the constitution; the same ε pays both, at different instants.

0.4 — Epistemic status per section

§2 — Constitution record — DERIVED (definition-level, from R)

§3 — Held, not shed — DERIVED at the structural register; record-algebra formalisation owed (D-CCC.1); guarded by KS-CCC.5

§4 — One arena, one snap, one ε — DERIVED (the clause in its native scope + AP43 D5); Corollary DERIVED (application of AP29)

§5 — Realised over provisioned — Form DERIVED (read off the axiom); value of ε SELECTED AMONG STRUCTURAL CANDIDATES (D-CCC.4)

§6 — Placement — DERIVED — forced under the arena reading; two-candidate elimination retained as corroboration

§7 — Double entry — STRUCTURAL — EXTENSION of AP24's licence, stated as such; guarded by KS-CCC.6

§8 — Vertices — DERIVED count; MOTIVATED assignment (D-CCC.2); empirically silent

§9 — Prediction — EMPIRICAL, reported as non-exclusion (KS-CCC.3)

§10 — The rule — METHOD (KS-CCC.1)

§11 — Candidate three — STRUCTURAL PLACEMENT (KS-CCC.4 switch)

0.5 — Kill switch summary

Seven kill switches engage, identifiers pending registration in the Master Kill Switch Registry. **KS-CCC.1** (forced structure): LIVE — METHODOLOGICAL. **KS-CCC.2** (mass-side drift): LIVE — DISCIPLINE. **KS-CCC.3** (existence, direction, and the migration fork): LIVE — EMPIRICAL. **KS-CCC.4a** (gap-scale exclusion) and **KS-CCC.4b** (scatter structure): LIVE — EMPIRICAL. **KS-CCC.5** (held, not shed): LIVE — STRUCTURAL. **KS-CCC.6** (single entry): LIVE — STRUCTURAL, coupled to KS-R.9.

Stratified honestly, so no reader has to sort the pile: only KS-CCC.3 and KS-CCC.4a/4b can be fired by nature. KS-CCC.1 binds the claim. KS-CCC.2 is, under §4's corollary, a discipline fence rather than an empirical test — its category changed when the corollary made commitments constitutive of mass, and the label now says so. KS-CCC.5 asks for a structural counter-argument; KS-CCC.6 waits on another paper's debt. The theory's live empirical content is two tests: the direction of metrological migration, and the scale of apparatus corrections.

0.6 — Structural debts owed

D-CCC.1 (Formalisation). The held commitment is derived by contradiction at the structural register (§3); its formal expression as an operator on the record algebra is owed. Load-bearing for the paper's grade, named rather than hidden.

D-CCC.2 (Vertex uniqueness). The assignment of the commitment to the ten vertices is motivated, not forced — the mirror of KS-30.4's uniqueness debt. Non-load-bearing: §4 fixes the total independently of any distribution.

D-CCC.3 (Value audit). The realised coupling shifts G and the holding limit κ . Owed before adoption: an audit of every corpus locus that consumes the predicted value of G or κ numerically — not only formula-level dependents of α_G , but any passage computing a number from the predicted G . Formulas and values both.

D-CCC.4 (Boundary selection). The value of ε in the quotient is α_{em} — a selection among the domain readings AP06 §0.7 holds open, argued in §5 from

the coupling denomination, not derived. The formal closure of the selection is owed. Load-bearing for §9: if the constitution reads at a different boundary, the correction rescales and the prediction fails.

0.7 — Items held open without claiming debt status

The attribution of the -0.036% residual (§9) — laboratory convergence, higher order in ε , or a refinement not yet forced — is held open with no selection made. The difference between the quotient form and its first-order shadow (about 53 ppm in G) and the difference between distributed and undistributed vertex forms (about 24 ppm) are both below current decidability and are held open as stated. Two prices of the arena corollary are named rather than left for discovery: Proposition 2 applied to matter configurations has no observable consequence and no switch — an unfenced universal, the cost of trivial ratio invariance; and the commitment's entire empirical content is one factor in one constant. AP30's decomposition is henceforth read as commitment-inclusive, with no layer identifying the commitment — a new reading at zero numerical cost, logged for the adoption table. The adoption sequence — the coordinated update table across AP28, Ø Predictions, the Proofs and the Prose — is owed on adoption, not before. The provenance records PROV-CCC.1 through PROV-CCC.5, the two session records, the transcript-export artifacts, and the analog photograph archive are published beside this draft.

0.8 — Structural relationships

This paper completes a pairing the corpus began: AP06 prices what Axiom C exacts at every instant; this paper prices what Axiom R exacted once, at the instant there first was anything to exact it from. The two costs are one mechanism read at two instants — the ε of a constitution instant is a commitment; the ε of every later instant is the leakage. Within Notebook IV, AP28 keeps the architecture (channels, puncture, compounding) untouched; this paper supplies the one factor the architecture provisioned and never booked. And AP29 closes the loop from the matter side: what each configuration holds is already on its books, as mass. Only the arena's account was open. This paper closes it.

1 — Definitions

The snap, and The Snap. A snap is a constitution event: the single Actualization State instant at which a configuration’s first record is written and holds. Before it there is possibility; after it there is a configuration paying maintenance. Every persisting configuration had one. The Snap, capitalised, is the arena’s own: the original break, the first record of all — the one constitution event with no arena prior to it. This paper prices The Snap, and §4 shows why the others are already priced.

Provisioned and realised. The axiom names two totals, and this paper uses both. What a snap requires is the full right-hand side of the axiom: $1:1 + 1 \times \varepsilon$ — the balance, plus the one element that holds it. What persists is the 1:1. “Provisioned” is the first total; “realised” is the second.

The commitment (CCC). The unpaired ε of the arena’s constitution instant, held as the price of the arena’s persistence. Paid once. Held forever. Denominated in the persistence coupling — it appears as a fraction of α_G , which is why G is the denominator of the configuration-cost estimate, as c is the denominator of every leakage estimate.

The leakage (TLC). The unpaired ε of every subsequent instant, shed at the boundary. Paid at every now. AP06’s subject, restated here only for the contrast:

	CCC — the commitment	TLC — the leakage
Paid	Once, at the arena’s constitution	At every now
Axiom	R — persistence	C — constraint
Geometry	Vertices of the coupling graph (10)	Edges of the coupling graph (21)
Denominator	G — the persistence coupling	c — the propagation bound
Where it appears	$\alpha_{G*} = \alpha_G / (1 + \varepsilon)$	Every boundary (AP06, Theorem 3.1)
Corpus home	This paper	AP06

Fix each cell of the table in mind and the rest of the paper is bookkeeping: one ε per instant, two destinations, and the destination is decided by whether the instant is a configuration’s first.

2 — The Constitution Record

Proposition 1 — The Constitution Record.

Statement. Every persisting configuration has a unique first record: the record that constitutes it.

Proof. Axiom R orders records irreversibly (AP01, AP20): for any configuration, its records stand in a strict temporal order that cannot be unwritten. A persisting configuration has at least one record — persistence is exactly the holding-open of a record; a configuration with none has nothing held and is not a configuration. A non-empty, strictly ordered, irreversible set of records has a least element, and strictness makes it unique. That least element is the constitution record: the writing of the configuration into the arena. ■

Applied to the arena itself, the proposition names the first record of all: the break. The arena is the one configuration whose constitution record has no arena prior to it — which is why §4 will find that its account, alone, was never booked.

3 — Held, Not Shed

This is the load-bearing section. Everything numerical in §9 hangs on it, and its grade is stated exactly: derived at the structural register, with the record-algebra formalisation owed as D-CCC.1 and the claim guarded by KS-CCC.5.

Proposition 2 — The Held Commitment.

Statement. At a configuration's constitution instant, the unpaired ε of that instant is held by the configuration as its persistence commitment. At every subsequent instant, the unpaired ε is shed — the leakage of AP06.

Proof. Step 1. Every Actualization State instant produces exactly one unpaired ε that does not return to symmetry (AP43 D5, carried into Notebook IV by AP06 §6). This holds at a constitution instant as at every other: the instant's inventory is the axiom's inventory, $1:1 + 1 \times \varepsilon$.

Step 2. A configuration persists only if its constitution record is held open. This is Axiom R read as AP28 §2 reads it: "Without persistence, the crack heals, the symmetry restores, records cease." Holding is not free-standing — something present must do it.

Step 3. Enumerate what is present at the constitution instant. The axiom's complete content at any instant is the $1:1$ and the $1 \times \varepsilon$ — there is nothing else to enumerate, and at the arena's own constitution the enumeration is airtight, because nothing else yet exists to be smuggled in. The $1:1$ is not committable: AP10 §7 identifies the $1:1$ as the substrate itself, the fifth degree of freedom, the probability dimension on which every configuration stands. Committing the stage to hold one record would not constitute a configuration; it would end the arena in which configurations occur. The only committable element present at a constitution instant is that instant's $1 \times \varepsilon$.

Step 4. Suppose, for contradiction, that the constitution ε is shed. Then by Step 3 nothing committable remains at the instant to hold the constitution record; by Step 2 the record is not held; by Axiom R the crack heals and the configuration does not persist — contradicting the supposition that it is a persisting configuration. Therefore the constitution ε is held.

Step 5. At every subsequent instant the constitution record already stands held: the commitment is in place, and the instant's ε has no constitutive work to do. It is shed at the boundary. That shedding, instant after instant, is precisely the leakage AP06 measures — thermal, boundary-informed, C-bounded. ■

One mechanism, two destinations. The dichotomy the corpus needed — a once-cost beside a flow-cost — does not require two mechanisms. It requires one: an ε

per instant, with the destination decided by whether the record it meets is being written or being held. The ε that meets a record being written becomes a commitment. Every ε after it becomes the leakage. The commitment is why the configuration exists; the leakage is what its existing costs the boundary from then on.

[A note held outside the load-bearing text. The Lock identifies the corpus's one ε with the electron. At the Lock's reading, then, the element the arena committed is the electron — and the old line about the electron as cost-bearer survives at that level: as a reading, not as a mechanism. Nothing in this paper rests on it.]

The grade, exactly. The contradiction closes at the structural register: every step cites a locked result, and the exhaustiveness step (Step 3) rests on AP10 §7 rather than on assertion — and is strongest precisely where the paper needs it, at the arena's own constitution. What is owed is the formal expression — the held commitment as an operator on the record algebra, doing in symbols what Step 4 does in structure. That is D-CCC.1, and it is this paper's standing debt in the sense that KS-30.1 is AP30's: the argument is strong, named, and not yet a theorem.

4 — One Arena, One Snap, One ε

The question this section answers is the one a careful reader asks first: every persisting configuration holds a commitment, and there are on the order of 10^{80} configurations — so why does the coupling carry one factor of $(1 + \varepsilon)$ rather than $(1 + \varepsilon)$ to some enormous power? The answer is that the exponent counts arenas, not configurations, and there is one arena.

Proposition 3 — One Snap, One ε .

Statement. The arena's commitment is exactly one ε .

Proof. Two steps, from two sources, kept separate. First, the arena constitutes entire: AP28 §4's clause is arena-scoped in its own words — “Axiom R does not permit selective persistence. The crack is open everywhere or nowhere.” The arena does not come into being region by region or channel by channel; its constitution is a single event, and Proposition 1 gives that event a unique record: the break. Second, one event carries one ε : an Actualization State instant produces exactly one unpaired element (AP43 D5). One arena, one snap, one instant, one ε . By Proposition 2, held. ■

Corollary — Already Booked. Matter configurations hold constitution commitments of their own, by the same Proposition 2 — and every one of them is already on the books. AP29 establishes $E \equiv \kappa(B) \cdot AS$: mass is held coupling capacity, actualised. A configuration's held ε is held coupling capacity; therefore it is a constituent of the configuration's mass — and the masses the corpus uses are measured masses, which contain everything the configuration holds. A configuration's commitment is not a correction waiting to be applied to its mass. It is part of what its mass has been all along. Nothing dresses any mass; no mass ratio can carry a commitment signature; the proton-to-electron ratio of AP30 is invariant trivially — not by cancellation, but because there was never anything to cancel — and AP30 §7's reading of its own residual, “It is not a missing term,” stands exactly as locked. The one commitment in the universe that no measured quantity contains is the arena's, because the arena is the one configuration that is not a mass inside itself. Its account alone was open. ■

This corollary is guarded by KS-CCC.2, kept deliberately as a fence against any future mass-side drift, and the guard is unforgiving for a reason worth restating: a commitment that scaled with a configuration's channel count would shift the proton-to-electron ratio at eight orders of magnitude above what measurement allows. The arena reading does not merely survive that constraint. It explains why the constraint was never in danger: matter's commitments were priced into matter from the start.

5 — Realised over Provisioned

Proposition 4 — The Form.

Statement. The realised coupling is the structural coupling divided by $(1 + \varepsilon)$: $\alpha_G^* = \alpha_G / (1 + \varepsilon)$.

Proof. The axiom names both totals. What The Snap provisioned is its full right-hand side: $1:1 + 1 \times \varepsilon$ — the balance, plus the one element whose holding makes the balance persist (Propositions 2 and 3). What persists as the realised arena is the 1:1. The realised fraction of the provisioned is therefore $1 / (1 + \varepsilon)$: the balance over the balance-plus-commitment. The correction is a quotient because the axiom is a sum. ■

The first-order shadow $(1 - \varepsilon)$ differs from the quotient at order ε^2 — about 53 parts per million in G — below anything current metrology can resolve. The structural form is the quotient, because the quotient is what the axiom writes, and every number in this paper is computed in the quotient convention.

The value of ε in the quotient — a selection, argued and booked. The form above is derived; the value is not, and the paper says so. AP06 §0.7 holds open a family of domain readings of ε — the leakage ratio at the gravitational boundary near 10^{-17} , at the retinal boundary between 10^{-6} and 10^{-5} , at the cosmological boundary near 10^{-5} — beside the corpus-wide $\varepsilon = \alpha_{em}$ at the coupling boundary. Choosing among them is a selection, and it is graded here as AP28 grades its $1/\pi$: SELECTED AMONG STRUCTURAL CANDIDATES, with the argument stated and the closure owed (D-CCC.4).

The argument for the coupling denomination. The commitment is a fraction of the provisioned coupling — a discount is denominated in the currency of what it discounts, and the currency of α_G is fixed by AP28's own Proposition 5: α_{em} is, by definition there, the unpaired element's one-channel coupling worth. One committed element, read in coupling currency, is one α_{em} . The domain values AP06 catalogues are of a different kind — they are leakage ratios of specific absorbing systems, flow quantities measured at boundaries, not the element's denomination in the coupling it discounts. The selection is therefore argued, not arbitrary — and not derived. If the constitution is shown to read at a different boundary, the correction rescales and §9 fails. That sentence is the honest price of this section, and D-CCC.4 keeps it visible.

6 — The Placement

Proposition 5 — The Arena’s Commitment Lives in the Arena’s Coupling.

Statement. The commitment divides the persistence coupling α_G , and can appear nowhere else.

Argument. Under the arena reading the placement is forced, not selected. The commitment is the arena’s (§4). The corpus has already named the quantity that is the arena’s own cost of persisting — AP28 §2, locked: “ $G \leftrightarrow$ Axiom R (Record — persistence). The cost the arena pays to keep the break open.” The arena’s commitment can only appear in the arena’s coupling, and the arena’s coupling is G . There is no second candidate to eliminate: matter’s commitments are already booked as mass (§4, Corollary), so no mass is available to dress, and no other universal quantity is the arena’s persistence cost by the corpus’s own mapping. $\alpha_{G^*} = \alpha_G / (1 + \varepsilon)$; $\kappa^* = \alpha_{G^*} \hbar c / 2$. ■

The elimination that a weaker reading would have needed is retained as corroboration, because it is independently sound: even if the commitment were sought on the mass side, that home fails twice — the arithmetic requires an unforced coefficient of one half to pass through m_e^2 , and AP29 makes the booking a double count, since the held capacity is already the mass. The forced placement and the failed alternative agree, which is what completeness feels like when it arrives.

The placement fixes the paper’s footprint, and the footprint is the proposal’s discipline. Mass ratios: untouched, trivially (§4). The acceleration floor $a_0 = C_S^2 \cdot cH_0 / (2\pi)$, the Hubble prediction, and the dark-sector fractions: untouched at the formula level — none contains α_G . The nuclear binding ceiling of AP30 §8: untouched — it carries the puncture factor, not the compounded coupling. What moves: the predicted G , and the holding limit κ . And the audit this forces is wider than formulas: any corpus passage that consumes the predicted value of G or κ numerically — a worked example, a table, a Planck-scale estimate — moves with it, even where no α_G appears in the expression. D-CCC.3 books the full value-level audit, owed before adoption.

7 — The Double Entry

Proposition 6 — Two Readings, One Element: an Extension, Stated as an Extension.

Statement. The puncture factor $(1 + 1/\pi)$ and the commitment quotient $1/(1 + \varepsilon)$ are two independent readings of the one unpaired element — what it contributes and what it committed — and their joint appearance in the realised formula is not double-counting.

Argument. Be precise about the licence. AP24 licenses reading one ε through different instruments — as G, as c, as α_{em} : across instruments. This section needs more: two readings of the element inside one instrument, multiplicatively, in α_G — and α_{em} itself already plays the per-channel role twenty-one times in the same expression. That is an extension of the licence, and the paper claims it as one, not as a straight application.

What the extension requires, and what supports it. It requires that “what the element contributes” and “what the element committed” be independent questions even at a single boundary — that contribution and cost not be one projection wearing two names. The support is structural, from §8’s graph: the puncture factor is an edge-stratum statement — the capacity the element’s round hole adds to a coupling channel (AP28, Proposition 4) — while the commitment is a vertex-stratum statement — the provision the element’s constitution required (§3). The two within-formula readings are readings of different strata of one proven structure, which is as close as the corpus currently comes to within-instrument independence. Close is not closed: the extension is claimed, supported, and guarded — not proved. And the dependency is declared rather than hidden: §8 is load-bearing for this section’s support — for the independence of the two readings — and for nothing arithmetic; the MOTIVATED grade of §8’s assignment is inherited here as the grade of the support. ■

The guard is KS-CCC.6, and it is coupled to a debt the corpus already carries. AP28’s KS-R.9 sub-debt owes the formal normalisation of the boundary coupling. If that calculation, when completed, is shown to already contain the persistence provision — one reading, not two — then the commitment double-counts, the correction is removed, and the puncture factor absorbs it under a reinterpreted normalisation. The switch is stated so that the hostile reading of this section — “first a factor was added to fit, now a factor is divided to fit better” — meets its answer in the register where it belongs: a named test, not a rhetorical defence.

8 — The Vertices

The count is already proven; only the reading is new. AP28 Propositions 1–2 establish the coupling structure of the arena: six faces of ε , three spatial dimensions, one Actualization State. Faces couple to dimensions — eighteen edges. The Actualization State couples to dimensions — three edges. Faces do not couple to the Actualization State — Proposition 2, the page and the pen. Twenty-one edges. And the same proven structure has $6 + 3 + 1 =$ **ten vertices**.

The two costs of §1 are the two counts of this one graph. Leakage is transmission — an edge phenomenon: it flows along the couplings, per instant, C-bounded, twenty-one wide. Constitution is existence — a vertex phenomenon: the configuration is written into the elements themselves, once, R-recorded, ten deep. Under the arena reading the fit is exact: The Snap is what constituted the graph itself, and the commitment is written into the ten vertices at the instant they first exist. The corpus's ten stands to its twenty-one as node to edge, as constitution to flow, as once to always. Nothing was added to obtain the ten. It was standing in AP28's proof the whole time, waiting for the question that counts vertices instead of edges.

Honesty about status, in two halves. The count is DERIVED — it is the vertex set of a proven graph. The assignment — the commitment distributed one share per vertex, compounding to the quotient of §5 — is MOTIVATED, not forced, and it is booked as such (D-CCC.2), the exact mirror of KS-30.4's uniqueness debt in AP30. And the assignment is empirically silent: the distributed and undistributed forms differ by about 24 parts per million in G, below anything the laboratories can currently see. Because §4 fixes the total at one ε regardless of any distribution, nothing numerical in this paper rests on the vertices. They are architecture — the right architecture, this paper holds — and architecture is claimed as architecture, not smuggled in as arithmetic.

9 — The Prediction

The realised gravitational coupling is $\alpha_{G*} = \alpha_{em}^{21} \times (1 + 1/\pi) / (1 + \alpha_{em})$, and the realised constant is:

$$G = \alpha_{em}^{21} \times (1 + 1/\pi) / (1 + \alpha_{em}) \times \hbar c / m_e^2$$

Using the CODATA inputs, as AP28 uses them:

$$\alpha_{em}^{21} = 1.338 \times 10^{-45}$$

$$1 + 1/\pi = 1.31831$$

$$1 / (1 + \alpha_{em}) = 0.992756$$

$$\text{Structural value (AP28): } G = 6.7206 \times 10^{-11} \text{ N}\cdot\text{m}^2/\text{kg}^2$$

$$\text{Realised value: } G = 6.6719 \times 10^{-11} \text{ N}\cdot\text{m}^2/\text{kg}^2$$

$$\text{Measured (CODATA 2018, unchanged 2022): } G = 6.6743 \times 10^{-11} \text{ N}\cdot\text{m}^2/\text{kg}^2$$

Residual: -0.036%.

Read against the landscape rather than against a single number. AP28's structural value stands 0.69% above the recommended value — thirteen times the roughly 500 ppm spread between independent laboratory determinations, outside every experiment ever performed. The realised value stands about 360 ppm below the recommended value — inside that spread, within one standard uncertainty of the lowest published determination and below others. The commitment closes about ninety-five per cent of the discrepancy and moves the prediction from outside the experimental landscape to its low edge. That is what the correction does. And stated against individual experiments, as a hostile reader will state it first: the realised value sits twenty-nine standard deviations below HUST-T-18 and twenty-five below UWash-00 in those experiments' own uncertainties, and sixteen below the CODATA recommendation at its formal 22 ppm. The commitment moved the exclusion from roughly six hundred standard deviations to twenty-nine — a twenty-fold improvement that is still exclusion, not agreement, and is claimed as nothing more.

And here is what the landing is, exactly: non-exclusion, not evidence. By §10's own argument the target is soft — an admissible window seven and a half per cent of ε wide against a 22 ppm formal uncertainty — and a realised value sitting essentially on top of the lowest modern determination is precisely the kind of agreement this paper declines to lean on. The prediction earns its keep from the fork it creates and the switches that guard it, not from where it happens to land today.

The sign flip, stated plainly. Under AP28 alone, the gap ran prediction-above-measurement, the direction downward apparatus leakage can explain — Ø Predictions Part II says exactly this. Under the commitment, the residual runs prediction-below-measurement, and the same leakage mechanism cannot absorb it: leakage lowers what laboratories read; it cannot raise it. Adopting this paper therefore trades a large residual with a comfortable story for a small residual with no story yet. The residual is reported open and unattributed (§0.7): laboratory convergence toward the low determinations, a higher-order term in ε , or a refinement not yet forced. Which of these, the data will say — under the switches of §12 — and this paper will not say first.

10 — What the Data May and May Not Do

The rule is already corpus precedent, and this section only names it. In AP28, structure proposed three candidate normalisations — $1/\pi$, $1/2\pi$, $1/4\pi$ — and measurement eliminated two. The direction of that traffic is the entire discipline: **data may eliminate structural candidates; data may never generate them.** A number that arrives from the measurement side, however clean it looks, is a fitted parameter wearing a costume, and the corpus's zero-fitted-parameters claim dies at the first one admitted.

Applied here, the rule has teeth, because the whisper is real and it is fenced by being named. All figures in this section are in the paper's structural convention, the quotient of §5. The commitment the recommended central value would demand, computed backwards, is 0.951 of one ε . Two clean fractions sit inside the window around it: **20/21 of ε lands about 12 ppm below the centre**, and **17/18 of ε about 45 ppm above it**. Both arrived from the data. Neither has a forcing argument. Both come pre-dressed in a story — an exempt channel here, a refunded share there — and the stories are exactly as good as the fractions, which is to say: generated by the answer. They are recorded in this section for one purpose. Seen. Refused. (In the discarded first-order shadow convention the backwards fraction reads 0.944 instead of 0.951 — and that the demanded fraction changes with a bookkeeping convention by an amount comparable to the gap between the whispers themselves (0.0066 of ε against their 0.0079) is one more proof that the digits cannot carry a decision.)

And the target itself is soft, which is the final proof that digit-matching is meaningless here. Measured against the lowest published determination the demanded fraction is 1.000 of ε — the paper's own magnitude, a landing §9 already declines to lean on; against the highest, 0.925; the admissible window is the whole range between, seven and a half per cent of ε wide against a formal uncertainty of 22 ppm. Any fraction in the window "fits". Structure alone can choose, and structure already chose before the window was drawn: the axiom writes one element, once. This paper states $1 \times \varepsilon$, states nothing finer, and arms KS-CCC.1 over every future refinement — whoever writes it — that cannot show its forcing argument was written before its number.

And the fence holds the paper's own history, because the selection pressure in this work did not run only on fractions — it ran on structure. Per-channel compounding and a channel-scaled cost were eliminated by measurement (legal under the rule, and disclosed); the count moved from ten-as-divisor to per-configuration inside the same measurement-laden opening analysis, and from per-configuration to the arena under purely structural pressure, with zero

numerical motion after the first computation; and the $1 \times \varepsilon$ magnitude argument was articulated in the same breath as the numbers, against a residual that was already published canon — and the honest genre carries two clauses, both true and both claimed: in process, a structural correction with refused tuning — the commitment was developed over weeks aimed at the proton and the neutron, blind to what it would do to G, and moved here only when AP30's precision closed the original target; in standing, postdictive — the 0.69% was published canon before this work began, and a landing on a known number is claimed as a landing, nothing more. The process is evidenced in PROV-CCC.1 through PROV-CCC.6; the standing is priced in §9. The dated sequence, with every elimination and its driver, is published as PROV-CCC.1 beside this draft, so that the claim of zero fitted parameters can be audited as a process rather than believed as a sentence.

11 — Candidate Three

Ø Predictions Part II holds the 0.69% open between two candidates and declines to pick: Candidate 1, apparatus leakage — the structural value is the unleaky value and every real measurement reads low; Candidate 2, refinement of the puncture normalisation under the KS-R.9 sub-debt. This paper enters as Candidate 3: the gap is neither in the apparatus nor in the puncture but in an unbooked provision — the structural value is the provisioned coupling, and what laboratories measure is the realised one.

Candidate 3 does not merely join the list; it resolves an inconsistency the corpus is already carrying, and the inconsistency should be stated in internal form. The corpus's own position — AP06, carried into AP28 §7 and Ø Predictions — is that the inter-laboratory scatter *is* the leakage signature. A signature sizes its effect. The scatter is about 500 parts per million; therefore, by the corpus's own identification, apparatus leakage is a 500 ppm effect. The gap is about 6,900 ppm. The corpus cannot simultaneously identify the scatter as leakage's signature and ask leakage to carry fourteen times its signature. Candidate 1, as currently written, does both. Candidate 3 dissolves the inconsistency by division of labour: a universal $1/(1 + \varepsilon)$ carries the gap, and leakage carries what its own signature says it carries — the scatter, and a residual at scatter scale.

The discriminator already exists in the corpus. Appendix C of Ø Predictions proposes the meta-analysis: characterise each published G experiment by its boundary geometry and test the leakage correlation. Under Candidate 1, the boundary-geometry corrections must come out near 6,900 ppm and correlate. Under Candidate 3, they must come out at scatter scale — hundreds of ppm at most. Same proposed experiment, opposite predictions, and KS-CCC.4 binds this paper to the outcome. Candidate 3 adds a falsification surface, not an escape hatch — which is the only way a third candidate earns entry into a register that was honest with two.

On Candidate 2, this paper is indifferent at entry and constraining at exit — by sign before magnitude. A commitment is a cost, and a cost can only reduce a prediction; it cannot raise one. The bare $1/2\pi$ and $1/4\pi$ normalisations give 5.9092×10^{-11} and 5.5036×10^{-11} — both already *below* the measured value. No commitment of any magnitude can rescue either; reaching measurement from there would require a refund — formally, commitments of -15.7ε and -24.0ε — and no reading of the axiom provides one. With the commitment live, $1/\pi$ remains uniquely selected among the enumerated candidates, now excluded by direction before arithmetic. The KS-R.9 sub-debt is untouched, and KS-CCC.6 states what happens if its resolution collides with §7.

12 — Kill Switches

Seven switches, Claim / Test / Status / Recovery, identifiers pending MKSR registration.

KS-CCC.1 — Forced structure

Claim. Every structural choice entering the commitment — count, placement, boundary, magnitude — is either forced from the axiom and locked results, or is a disclosed selection with its eliminations on the public record (PROV-CCC.1–6). This switch binds the corpus’s claim, not its author. The author is bound by nothing here — not this paper, not this corpus, not this switch: exploration, revision, and new derivation remain his without condition, the proton, neutron, and deuteron included. What the fence holds is the flag: the zero-fitted-parameters status of the G result.

Test. The standing in canon of any refinement of the commitment whose selection depended on comparison with measured G and whose selection history is undisclosed.

Status. LIVE — METHODOLOGICAL. It binds the claim.

Recovery. The flag retires for the refined result — it may stand as work, but not as parameter-free. The original $1 \times \varepsilon$ result, with its disclosed history, is unaffected.

KS-CCC.2 — Mass-side drift

Claim. No mass is dressed by the commitment: matter configurations’ commitments are already booked as mass (AP29; §4, Corollary), and m_p/m_e carries no commitment signature above experimental uncertainty (± 0.0174 ppb, CODATA 2022).

Test. Any derivation, anywhere in the corpus, that applies a commitment factor to a mass or exhibits a commitment residue in a mass ratio.

Status. LIVE — DISCIPLINE.

Recovery. The arena reading fails, and with it the paper: a mass-side or channel-scaled replacement is excluded by AP30 at eight orders of magnitude, so the proposal reverts to unpriced — AP28 §2’s sentence stands, the commitment withdrawn.

KS-CCC.3 — Existence, direction, and the migration fork

Claim. A commitment of order ε exists; the structural value exceeds the realised value; and — the fork — as apparatus systematics are resolved, the adjusted (consensus) value of G migrates downward, by roughly 300 ppm from its current centre, toward the realised 6.672×10^{-11} . The null (no commitment; AP28 coincidental) predicts it remains near 6.6743. This is the one forward prediction that separates the commitment from the null, and it is slow, dated, and out of the author's hands.

Test. Successive CODATA adjustments and replication-grade determinations, beginning with the first datum after this switch was written: NIST-26, which moved downward. Convergence upward toward 6.7206×10^{-11} kills the commitment and revives Candidate 1 in full. Migration downward toward 6.672×10^{-11} supports the commitment against the null; stability at 6.6743 as systematics close supports the null. Individual determinations may sit on either side of the realised value within their uncertainties (LENS-14's centre sits marginally below it); the fork is about the adjusted value, not any single experiment.

Status. LIVE — EMPIRICAL. The switch tests existence and direction, not magnitude: by §10, the admissible window is seven and a half per cent of ε wide, and convergence cannot distinguish $1 \times \varepsilon$ from nearby fractions. No claim of magnitude confirmation will be made from it.

Recovery. If metrology converges toward the structural value, the commitment is withdrawn and the 0.69% returns, whole, to Candidates 1 and 2.

KS-CCC.4a — Gap-scale exclusion

Claim. Apparatus corrections are not of gap scale: no physical characterisation of the published apparatuses supports a common leakage term of order 6,900 ppm.

Test. The Appendix C meta-analysis — executed twice at Tier 1, independently, on the same day (WP-CCC.1; the reviewer-thread execution), outcome N in both — and Tier 2's blind coding of absolute leakage capacity.

Status. LIVE — EMPIRICAL. Tier 1 did not fire it; the pass is exclusion of Candidate 1's scale, not confirmation of the commitment.

Recovery. Gap-scale corrections confirm Candidate 1 and withdraw the commitment; the realised formula reverts to AP28's.

KS-CCC.4b — Scatter structure

Claim. Residual boundary-dependence exists in the inter-laboratory scatter at AP06's honest size — order 10^2 ppm.

Test. Tier 2's blind boundary coding of the scatter, under a stated limit: implied leakage is affine in G , so class-correlation statistics are identical under every constant-true-value hypothesis — 4b tests AP06's residual-leakage reading of the scatter, and cannot by itself discriminate the candidates' central values.

Status. LIVE — EMPIRICAL. Tier 1 found the structure at best marginal ($p \approx 0.06-0.2$).

Recovery. If no boundary structure exists at any scale, AP06's reading of the scatter weakens; the commitment's central claim is unaffected either way.

KS-CCC.5 — Held, not shed

Claim. A constitution instant's ε is held as the persistence commitment (Proposition 2), and in particular the arena's constitution ε is held.

Test. Exhibit a mechanism by which a constitution ε — the arena's above all — is shed while the constitution record is held open by other means.

Status. LIVE — STRUCTURAL.

Recovery. Proposition 2 fails; the dichotomy collapses to leakage-only; the commitment is withdrawn. AP06 is unaffected — the leakage never depended on this paper.

KS-CCC.6 — Single entry

Claim. The puncture factor and the commitment are independent readings of the unpaired element (Proposition 6, claimed as an extension of AP24's licence).

Test. The completed puncture-topology normalisation owed under KS-R.9, if it is shown to already contain the persistence provision.

Status. LIVE — STRUCTURAL, coupled to KS-R.9.

Recovery. The correction double-counts and is removed; the puncture factor absorbs the commitment under a reinterpreted normalisation; the realised formula reverts to AP28's.

13 — Dependencies

Depends on: AP06 (Identification 6.1; the open domain family; the leakage as shed ε). AP28 (the coupling graph; Proposition 5's coupling denomination; the R-reading of G; the everywhere-or-nowhere clause). AP24 (the many-readings principle, extended in §7). AP10 (three dimensions; the 1:1 as substrate). AP43 D5 (one ε per instant). AP29 (mass as coupling capacity actualised — the booking of matter's commitments). AP20 (independence; the unconditional floor).

What depends on it: Nothing, until adoption. On adoption: AP28 §6–§7 and the wording of KS-R.7; Ø Predictions Part II (a third candidate; the sign flip of the residual; Appendix C read as the discriminator); the G passages of the Proofs (Chapters 16 and 19) and of the Prose and Metaphor registers; every corpus locus consuming the predicted value of G or of the holding limit κ numerically, per the widened D-CCC.3 audit — formulas and values both; and the Master Kill Switch Registry, which advances by seven switches; and AP30's decomposition, henceforth read as commitment-inclusive with no layer identifying the commitment. Adoption is a coordinated amendment sequenced by an update table, in the discipline the corpus already uses — not a patch, and not before the author's ruling.

Dated notes

11 September 2026. Six matters of record: five from the four papers locked this day (AP50–AP53), and one of dress. None alters the body above.

1. §2, Proposition 1: the premise is supplied. The break is the first record; grains pass one per click; the order is total and well-founded (AP50 §9).

2. §5 and §0.6: the value of ε moves from SELECTED AMONG STRUCTURAL CANDIDATES to READ BY RULE, three instances (AP50 §6). D-CCC.4 is closed on the rule; two instances meet it; the author: once would have been enough.

3. §5: the quotient's ground is restated as the leak booked against what is held (AP50 §7). The form is unchanged.

4. §7, Proposition 6, and KS-CCC.6: from extension to rule (AP50 §8). The coupling to KS-R.9 is released.

5. KS-CCC.1 is sharpened, as the Master Kill Switch Registry v5.31 prints it.

6. The republish of this paper removes one decorative character from the Artist's Note (U+2728), a matter of dress, recorded here and in the republish note.

Claim Summary

§2 — Constitution record. Every persisting configuration has a unique first record; the arena's is the break. DERIVED.

§3 — Held, not shed. A constitution instant's ε is held as the persistence commitment; every later instant's ε is shed as leakage. DERIVED — structural register; formalisation owed (D-CCC.1).

§4 — One arena, one snap, one ε . The exponent counts arenas; matter's commitments are already booked as mass (AP29). DERIVED; Corollary DERIVED; guarded by KS-CCC.2.

§5 — Realised over provisioned. $\alpha_G^* = \alpha_G / (1 + \varepsilon)$: form DERIVED; the value $\varepsilon = \alpha_{em}$ SELECTED AMONG STRUCTURAL CANDIDATES, argued from the coupling denomination, booked as D-CCC.4.

§6 — Placement. The arena's commitment lives in the arena's coupling — forced by AP28 §2's own mapping; the two-home elimination retained as corroboration. DERIVED.

§7 — Double entry. Contribution $(1 + 1/\pi)$ and commitment $1/(1 + \varepsilon)$: an extension of AP24's licence, stated as such, supported by the strata of §8's graph. STRUCTURAL — EXTENSION; guarded by KS-CCC.6.

§8 — Vertices. Ten vertices, twenty-one edges of AP28's proven graph; constitution at nodes, flow along edges. DERIVED count; MOTIVATED assignment; empirically silent.

§9 — Prediction. $G = 6.6719 \times 10^{-11}$; residual -0.036% , sign flipped, reported open; the landing reported as non-exclusion, not evidence. EMPIRICAL.

§10 — The rule. Data eliminates; data never generates. Demanded fraction 0.951 (quotient convention); 20/21 at -12 ppm and 17/18 at $+45$ ppm recorded and refused. METHOD.

§11 — Candidate three. The scatter-signature inconsistency stated in internal form and resolved by division of labour; the normalisations excluded by sign. STRUCTURAL PLACEMENT.

Conditionality Footer

Dependencies

AP06 (Identification 6.1; the open domain family). AP28 (coupling graph; Proposition 5; R-reading; everywhere-or-nowhere clause). AP24 (many-readings principle). AP10 (dimensions; the 1:1 as substrate). AP43 D5 (one ε per instant). AP29 ($E \equiv \kappa(B) \cdot AS$). AP20 (independence). This paper is unconditional on EH and QRA.

Dependents

None until adoption. On adoption: AP28 §6–§7, KS-R.7 wording, Ø Predictions Part II and Appendix C, Proofs Chapters 16 and 19, the Prose and Metaphor G passages, every value-level consumer of predicted G or κ (D-CCC.3), the MKSR.

Kill switches engaged

KS-CCC.1 (forced structure, LIVE — METHODOLOGICAL). KS-CCC.2 (mass-side drift, LIVE — DISCIPLINE). KS-CCC.3 (existence, direction, and the migration fork, LIVE — EMPIRICAL). KS-CCC.4a (gap-scale exclusion) and KS-CCC.4b (scatter structure), LIVE — EMPIRICAL. KS-CCC.5 (held not shed, LIVE — STRUCTURAL). KS-CCC.6 (single entry, LIVE — STRUCTURAL, coupled to KS-R.9).

Structural debts owed

D-CCC.1 (record-algebra formalisation of the held commitment — load-bearing for grade). D-CCC.2 (vertex-assignment uniqueness — non-load-bearing). D-CCC.3 (value-level audit of every consumer of predicted G or κ — owed before adoption). D-CCC.4 (boundary selection: the value $\varepsilon = \alpha_{em}$ in the quotient — load-bearing for §9).

Items held open without claiming debt status

Attribution of the -0.036% residual. The ε^2 -order difference between quotient and shadow (53 ppm). The empirical silence of the vertex distribution (24 ppm). The adoption update table, owed on adoption.

Notation Reference

1:1 + 1×ε @ AS — The axiom. The balance, the one unpaired element, read at the Actualization State.

ε — The unique unpaired element of Axiom B. Its value in the quotient, α_{em} , is a selection among AP06's open domain family (§5; D-CCC.4).

α_{em} — The fine-structure constant $\approx 1/137.036$. The per-channel coupling (AP28 Proposition 5) and the leakage rate (AP06).

α_G — The structural (provisioned) gravitational coupling, $\alpha_{em}^{21} \times (1 + 1/\pi)$ (AP28).

α_{G*} — The realised gravitational coupling, $\alpha_G / (1 + \varepsilon)$. Introduced in this paper.

κ, κ^* — The holding limit and its realised value, $\kappa^* = \alpha_{G*} \hbar c / 2$.

G — Newton's constant. Realised: $G = \alpha_{G*} \hbar c / m_e^2 = 6.6719 \times 10^{-11} \text{ N}\cdot\text{m}^2/\text{kg}^2$.

snap / The Snap — A constitution event; capitalised, the arena's own — the original break, the first record of all.

commitment (CCC) — The arena's constitution ε , held once as the price of the arena's persistence. G-denominated.

leakage (TLC) — Every subsequent instant's ε , shed at the boundary. c-denominated (AP06).

provisioned / realised — The axiom's two totals: 1:1 + 1×ε, and the persisting 1:1.

vertex / edge — The two counts of AP28's proven coupling graph: 6 + 3 + 1 = 10 vertices; 18 + 3 = 21 edges.

Actualization State (AS) — The actualising now: the surface from which records are written (AP01).

End of The Snap. Artist's Proof 44 — LOCKED 2026-08-11.

Catalogue	Ø Artist's Proofs
Proof Number	AP44
Title	The Snap
Subtitle	Forces and Constants
Medium	Natural Philosophy / Physics
Artist	G

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